[Final Exam (Sample)]

Spring 2024: CSE 309: Theory of Automata

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Sunday, May 26, 2024. Total Marks: 35 (final) + 5 (quiz) + bonus, Duration: 3 hours

Name: _____, Student ID: _____

Please read these instructions carefully:

- This exam contains 8 questions. Question 1 contributes towards the Quiz component of the course. The rest of the questions contribute towards the final exam component. Question 8 is a bonus question.
- Use of calculators, mobile, or any other communication devices is strictly prohibited for the entirety of the exam period.
- Please submit your devices in your bag at the front of the examination room.
- You may keep writing material with you on your desk.
- Read all questions carefully before you start answering and prioritize which questions to attempt first.
- You may ask for extra sheets for your rough work, scribbling, future planning etc., **anything on extra sheet(s) will not be graded.**
- Please write your name and student ID on this question paper clearly.
- Please do not use pencils or red/green/yellow pens for answers that you want to be graded.
- Instructor(s) will not entertain any queries during the exam.
- Please attach all extra sheets with your answer book.
- Please write clearly and legibly and avoid overwriting.
- Pay attention to the instructions for each question.
- Acquisition of answers through unfair means will automatically cancel your exam.
- Keep track of the time.
- **Your answer(s) will not be graded if unreadable (by the instructors)**.
- **Your exam will not be graded if you don't follow these instructions.**

— Good Luck —

Question 1: Quiz ..... *5 marks*

Consider the following pair of languages:

$$\textsc{Diag} := \{\langle M, k\rangle : M \text{ is a square matrix and sum of values on its diagonal is at most } k\}$$
$$\text{TSP} := \{\langle M, k\rangle : M \text{ is a square matrix representing inter-city distances and there is tour that visit all cities of total cost at most } k\}$$

Note that TSP is a known **NP**-hard problem, whereas DIAG is in **P**. Which of the following must be true, assuming $\mathbf{P} \neq \mathbf{NP}$? [No justification is needed.]

(a) ____۴____ TSP $\in$ **P**

(b) ____۲____ DIAG is in **NP**

(c) ____۲____ There is a polynomial-time reduction from SAT to TSP.

(d) ____۲____ There is a polynomial-time reduction from DIAG to TSP.

Question 2: Chapter 1 ..... *6 marks*

(a) (3 marks) Fix the alphabet $\Sigma = \{0, 1\}$. A *run* in a string $w \in \Sigma^*$ is a maximal substring of [redacted]. For example, the string `001110011` has exactly four runs: `001110011` = `00` · `111` · `00` · `11`.

Let L be the set of all strings in Σ^* that contains [redacted]. For example, L contains `00011111110` but does not contain the string `10000111`. Describe a regular expression for L.

Not sure is completely correct

Odd length runs mean that ε is not possible.

$L = [(0 \cap (00)^*)^+ \cap (1 \cap (11)^*)^+]^* \cup [(1 \cap (11)^*)^+ \cap (0 \cap (00)^*)^+]^*$

$(0, 000, 00000, \ldots)^* (1, 111, 11111, \ldots)^* \cup (1, 111, 1111 \ldots)(0, 000, 00000, \ldots)$

(b) (3 marks) Describe a CFG that generates the following language: Chapter 2

$$L = \{a^x b^y c^z : y = x + z \text{ and } x, y, z \geq 0\}$$

pretty sure mostly correct

$x=2, z=3, y=5$

$S \to A \mid \varepsilon$ // $x, y, z = 0$

$A \to A_2 C$

$A_2 \to aA_2b \mid \varepsilon$

$C \to bCc \mid \varepsilon$

$S \to A$
$A \to A_2 C$
$= aA_2bC$
aaA_2bbC
$aa\ \ bbC$
$= aabbC$
$= aabbbCcc$
$= aabbbbCccc$
$= aabbbbccc$
$= aabbbbbccc$

Question 3: ..... *6 marks*

Indicate whether following statements are **true** or **false** also justify your answer i.e., give a simple proof for a **true** statement and a counterexample for a **false** statement.

(a) True Every regular language is context-free. Chapter 2/2

It is true, proof is iffy

It is possible to convert all DFAs to CFGs, and all regular languages can be represented by DFAs for each, likewise all context-free languages can be represented by CFGs.

Hence, every regular language is context-free.

(b) _____ Following language L is **NP**-hard. Chapter 7

is false I think.

$$L = \{\langle M \rangle : M \text{ is a DFA that accepts all strings}\}$$

$L(M) = \Sigma^*$

Question 4: Chapter 5 ..... *5 marks*

Show that if A is *recognizable* and $A \leq_m \overline{A}$, then A is *decidable*.

pretty sure is (completly) correct

If $A \leq_m \bar{A}$, then by definition of the reducability function, $(\bar{A}) \leq_m (\bar{\bar{A}}) \rightarrow \bar{A} \leq_m A$.

Now, we are given that A is recognizable. From this, it follows that $\bar{A}$ must be recognizable, as $\bar{A}$ reduces to A.

Hence, is both A and $\bar{A}$ are recognizable, then it is known by a theorem that a problem is decidable iff it and its complement are recognizable. Since this is the case in this scenario, A is decidable.

Question 5: Chapter 7 ..... *6 marks*

Show that following problems are in class **NP**.

(a) **3SAT**: Given a CNF formula ϕ with n variables and m clauses, is there an assignment to the variables that satisfies all the clauses.

pretty sure correct

We can show that 3SAT $\in$ NP by creating a polynomial time verifier. Let that verifier be V. V takes in as input a CNF formula ϕ with n variables and m clauses, and a certificate c, which will include a a number of assignments for variables. // c is a set of assignments.

V = "On input $\{\langle\phi\rangle, c\}$:

1. Check if variables of c match variables in ϕ.
2. If check passed, then apply assignments to variables, else reject.
3. If ϕ evaluates to true, accept, else if it evals to false, reject.

(b) **Dominating Set:** Given a graph $G = (V, E)$ and a number k, is there a set $V' \subseteq V$ such that $|V'| \leq k$ and for every vertex $v \in V$, either v is in V' or v is adjacent to a vertex in V'.

pretty sure correct

Verifier (G, k, c)

We can prove that this problem $\in$ P by creating a polynomial time verifier for it. Let that verifier be V. V will take as its inputs $\langle G\rangle$, k, and a certificate c, which is a graph itself.

V = "On input $\{\langle G\rangle, k, c\}$:-

1. Verify that the vertices V' in C are less than k.
2. If the check passed, then verify the following for every $v \in V$ of graph $\langle G\rangle$:
   - is $v \in V'$, or
   - v is adjacent to a vertex in V'.
3. If both step 1 and step 2 is passed, then accept. Else, reject.

V runs in polynomial time, hence this problem $\in$ NP by the definition of the class of NP.

Question 6: Chapter 7 *5 marks*

If L is a language over some alphabet Σ^* then the complement of L denoted as $\overline{L}$ is defined as $\overline{L} = \Sigma^* - L$. We say a language L is in the class coNP if its complement $\overline{L}$ is the class NP. Prove/disprove that *"NP = coNP if and only if $A \leq_P B$ and $B \leq_P A$ where A is an NP-complete languages and B is a coNP-complete language."*

pretty sure correct, though not completed cause lazy

$L \in \overline{NP}$ iff $\overline{L} \in NP$

prove: $NP = \overline{NP}$ iff $A \leq_p B$ and $B \leq_p A$

- $A \in NP$-Complete
- $B \in \overline{NP}$-Complete

Proving Forward Direction:-

if $A \leq_p B$, and A is NP-Complete, then B must by definition also by NP-complete.

However we have $B \in \overline{NP}$-complete, and $B \leq_p A$, so by this, $A \in \overline{NP}$-Complete.

Clearly, these two points are contradictory, unless $NP = \overline{NP}$, and this is the only conclusion that can be reached.

Proving Backward Direction:-

If we have $NP = \overline{NP}$, then for any problems A and B $\in NP$, A and B $\in \overline{NP}$. Hence, if $A \leq_p B$, and we consider A to be

Question 7: Chapter 5 ..... *7 marks*

Let $T = \{\langle M \rangle \ M \text{ is a TM that accepts } w^{\text{rev}} \text{ whenever it accepts } w\}$. Prove by reduction that T is undecidable.

Please ensure that you give a complete argument regarding why your reduction works. You can choose any undecidable language that we have seen in class for your reduction.

lazy, the answer is of Anas.

Let us assume that T is decided by a TM H. We can reduce H to A_{TM}, hence $H \leq_m A_{TM}$. By the definition of the reducible function, A_{TM} will be decidable, since H is decidable. Let S be the TM that decides A_{TM}. To construct S, we can use TM H as part of the construction. TM S will take as input $\langle M, w \rangle$, where w is a string.

S = "on input $\langle M, w \rangle$:"

(1) Construct a TM M' based on M and w:

M' = "on input $\langle x \rangle$, a string:"

- If $x \neq w^{rev}$

Let T = { <M> | M is a TM that accepts w' whenever it accepts w }.
Assume T is decidable and let decider R decide T. Reduce from A_{TM} by constructing a TM S as follows:

- S: on input <M,w>
  1. create a TM Q as follows:
     On input x:
     1. if x does not have the form 01 or 10 reject.
     2. if x has the form 01, then accept.
     3. else (x has the form 10), Run M on w and accept if M accepts w.
  2. Run R on
  3. Accept if R accepts, reject if R rejects.

Because S decides A_{TM}, which is known to be undecidable, we then know that T is not decidable

Question 8: Bonus Chapter 7 ..... *5 marks*

Show that the class **NP** is closed under the star operation. That if L is in **NP** then L^* is also in **NP**.

this answer is correct I reckon

To show that $L^* \in NP$, we can create a polynomial time NTM M that takes as input a string $w \in L$. If $L^* \in NP$, then $w^* \in L^*$.

M = "On input w:
1. If $w = \epsilon$ then accept.
2. Nondeterministically select a number m such that $1 \leq m \leq |w|$.
3. Nondeterministically split w into m pieces such that $w = w_1 w_2 \ldots w_m$.
4. For all i, $1 \leq i \leq m$: run M_1 on w_i. If M_1 rejected then reject.
5. Else (M_1 accepted all w_i, $1 \leq i \leq m$), accept."

Observe that steps 1. and 2. take time $O(n)$, because the size of the number m is bounded by n (the length of the input). Step 3. is also doable in polynomial time (e.g. by nondeterministically inserting m separation symbols # into the input string w). In step 4. the for loop is run at most n times and every run takes at most $O(n^k)$. So the total running time is $O(n^{k+1})$. This means that M is a poly-time nondeterministic decider for L_1^*.